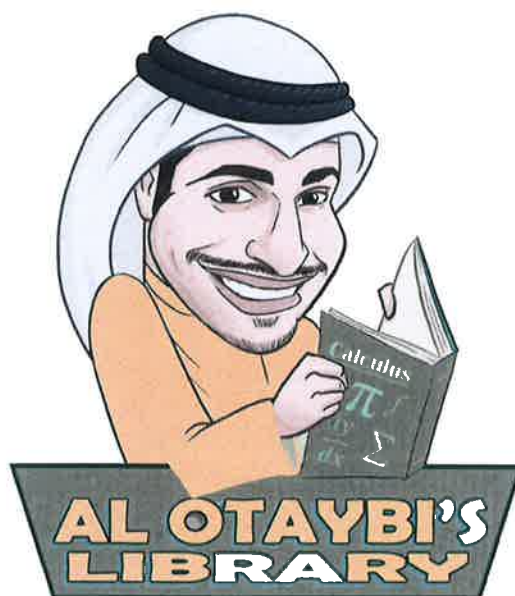


Kuwait University

2021/2022

Fall course



Calculus B

First

Section : 7.4 (Part a/c)

7.4 Integration of Rational Functions by Partial Fractions.

(6 a)



Calculus B Quiz # 5



$$\int \left[\frac{1}{x^2 + 5} + \frac{1}{x^2 - 5} \right] dx$$

$$\int \frac{x^3}{x^4 + 2} dx$$

$$\int \frac{\cos x}{\sin^2 x + 9} dx$$

$$\int \frac{e^x}{e^{2x} - 9} dx$$

$$\int \frac{x}{x^4 + 9} dx$$



$$\int \frac{x^4}{x^2 - 9} dx$$

أنواع التكاملات



By parts

عند وجود أولوية من الأولويات



u

dv

بـس دوال مثلثية

أو (h)

حدودية من
الدرجة الثانية

حدودية من
(الدرجة الثانية)

$\frac{1}{2} \frac{3}{2} \frac{5}{2} \dots$



AL OTAYBI'S
LIBRARY

حدودية
حدودية

المقام < البسط



Partial fraction

البسط $\leq$ المقام



قسمة مطولة



متفق عليه !!



.....
حدودية من الدرجة الثانية

حدودية من الدرجة الأولى
حدودية من الدرجة الثانية

ثابت
حدودية من الدرجة الثانية

يتم بناء الحل على
 $\ln$

يتم بناء الحل على
قاعدة السهم على المقام

$\tan^{-1}$

$\ln$



إذا كان المقام يتحلل يمكن حله

Partial fraction

$$\int \frac{2x + 1}{3x^2 + 3x + 10} dx$$

$$\int \frac{\frac{1}{3} (6x + 3)}{3x^2 + 3x + 10} dx$$

$$= \frac{1}{3} \ln |3x^2 + 3x + 10| + C$$

$$\int \frac{7}{x^2 + 2x + 10} dx$$

$$x^2 + 2x + 10$$

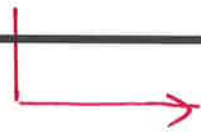
$$(x + 1)^2 - 1 + 10$$

$$(x + 1)^2 + 9$$

$$\int \frac{7}{(x+1)^2 + 9} dx$$

$$7 \cdot \frac{1}{3} \tan^{-1} \left(\frac{x+1}{3} \right) + C$$

$$\int \frac{8}{x^2 - 6x - 7} dx$$



$$x^2 - 6x - 7$$

$$(x - 3)^2 - 9 - 7$$

↓ * 1/2
- مربع

$$(x - 3)^2 - 16$$

$$\int \frac{8}{(x - 3)^2 - 16} dx$$

$$= 8 \frac{1}{2 \sqrt{16}} \ln \left| \frac{(x - 3) - 4}{(x - 3) + 4} \right| + C$$

(4)

$$= \cancel{8} \frac{\cancel{1}}{\cancel{8}} \ln \left| \frac{x - 7}{x + 1} \right| + C$$

$$\int \frac{x+1}{x^2+4x+7} dx$$

$$\int \frac{\frac{1}{2}(2x+4) - 1}{x^2+4x+7} dx$$

$$\int \left[\frac{\frac{1}{2}(2x+4)}{x^2+4x+7} + \frac{-1}{x^2+4x+7} \right] dx$$

$$\frac{1}{2} \ln|x^2+4x+7| + \int \frac{-1}{x^2+4x+7} dx$$

$$\downarrow \times \frac{1}{2}$$

$$(x+2)^2 - 4 + 7$$

$$(x+2)^2 + 3$$

$$\int \frac{-1}{(x+2)^2+3} dx$$

$$-\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x+2}{\sqrt{3}}\right) + C$$

$$\Rightarrow \frac{1}{2} \ln|x^2+4x+7| - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x+2}{\sqrt{3}}\right) + C$$

$$\int \frac{x^4 + 3}{x^2 + 5} dx$$

$$\begin{array}{r} \text{Divide } x^4 + 3 \text{ by } x^2 + 5 \\ \hline x^2 - 5 \overline{) x^4 + 3} \\ \underline{-(x^4 + 5x^2)} \\ -5x^2 + 3 \\ \underline{+ 5x^2 - 25} \\ 28 \end{array}$$

① ÷
 ② *
 ③ غير
 ④ +

$$\int \left[x^2 - 5 + \frac{28}{x^2 + 5} \right] dx$$

$$\frac{x^3}{3} - 5x + 28 \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{x}{\sqrt{5}}\right) + C$$

$$\int \frac{x^2 + x + 1}{x^2 - 1} dx$$

$$\begin{array}{r} -1 + x^2 \overline{) x^2 + x + 1} \\ \underline{-(x^2 - 1)} \\ x + 2 \end{array}$$

$$\int \left[1 + \frac{x+2}{x^2-1} \right] dx$$

$$x + \int \frac{\frac{1}{2}(2x) + 2}{x^2-1} dx$$

$$\int \left(\frac{\frac{1}{2}(2x)}{x^2-1} + \frac{2}{x^2-1} \right) dx$$

$$\frac{1}{2} \ln|x^2-1| + \frac{2}{2(1)} \ln \left| \frac{x-1}{x+1} \right| + C$$

$$\Rightarrow x + \frac{1}{2} \ln|x^2-1| + \ln \left| \frac{x-1}{x+1} \right| + C$$



Review



العامل المشترك



$$x^2 + 9x = x(x + 9)$$

$$3x^2 - 2x^3 = x^2(3 - 2x)$$

$$2x^2 - 4x = 2x(x - 2)$$

$$x - 9x^{10} = x(1 - 9x^9)$$



فرق بين مربعين



$$a^2 - b^2 = (a - b)(a + b)$$

$$x^2 - 9 = (x - 3)(x + 3)$$

$$4 - x^2 = (2 - x)(2 + x)$$

$$x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$$

$$4x^2 - 9 = (2x - 3)(2x + 3)$$

$$x^2 + 9$$



لا يمكن تحليلها



فرق أو مجموع مكعبين



$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$$

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

$$27x^3 - 8 = (3x - 2)(9x^2 + 6x + 4)$$





تحليل حدودية من الدرجة الثانية



$$ax^2 \pm bx \pm c$$



$$x^2 - x - 2$$



$$(x+1)(x-2)$$

$$x^2 - 5x - 6$$

$$(x+1)(x-6)$$

$$x^2 + 2x - 3$$

$$(x-1)(x+3)$$

$$x^2 - 6x - 7$$

$$(x+1)(x-7)$$

$$x^2 + 4x - 12$$

$$(x-2)(x+6)$$

$$x^2 + 7x + 6$$

$$(x+1)(x+6)$$

$$x^2 + 6x + 5$$

$$(x+1)(x+5)$$

$$x^2 - 4x + 3$$

$$(x-1)(x-3)$$

$$x^2 - 3x + 2$$

$$(x-1)(x-2)$$



$$2x^2 - 3x + 1$$



$$(2x-1)(x-1)$$

$$3x^2 - 7x + 2$$

$$(3x-1)(x-2)$$

$$3x^2 - 11x + 6$$

$$(3x-2)(x-3)$$

$$x^3 + x^2 - x - 1$$

$$x^2(x+1) - (x+1)$$

$$(x+1) [x^2 - 1]$$

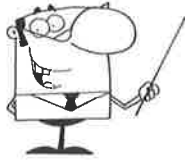
$$(x+1)(x-1)(x+1)$$

$$(x+1)^2(x-1)$$

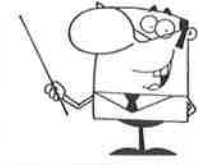
$$x^3 + 2x^2 + 2x + 4$$

$$x^2(x+2) + 2(x+2)$$

$$(x+2) [x^2 + 2]$$



Partial fraction



أسلوب يتم اللجوء له في حال عني

درجة المقام < درجة البسط

حدودية
حدودية

طريقة الحل

تسمى
Partial decomposition

(1) تحليل المقام الى أقصى درجة ممكنة
(2) توزيع المقام مع مراعاة البسط

$$\frac{\dots}{\dots} + \frac{\dots}{\dots} + \dots$$

(3) الضرب بالمقام للطرفين

(4) ايجاد المجاهيل

(1) التعويض بأصفار المقام

(2) التعويض بالصفر

(3) مساواة المعاملات Equate the coefficient

1 - المقام

ركاده!!!

قد يتم اجمال أحدها

$$\int \frac{x^2 + x + 1}{(x^2 + 2)(x - 1)} dx$$

$$\frac{x^2 + x + 1}{(x^2 + 2)(x - 1)} = \frac{Ax + B}{x^2 + 2} + \frac{C}{x - 1}$$

$$* (x^2 + 2)(x - 1)$$

$$x^2 + x + 1 = (Ax + B)(x - 1) + C(x^2 + 2)$$

$$\underline{x = 1}$$

$$1 + 1 + 1 = 0 + C(1 + 2)$$

$$3 = 3C \rightarrow \boxed{C = 1}$$

$$\underline{x = 0}$$

$$0 + 0 + 1 = (0 + B)(-1) + C(0 + 2)$$

$$1 = -B + 2C$$

$$1 = -B + 2(1) \rightarrow B = 2 - 1$$

$$\boxed{B = 1}$$

Equate the coeff of x^2

$$1 = A + C$$

$$1 = A + 1 \rightarrow \boxed{A = 0}$$

$$\int \left[\frac{0 + 1}{x^2 + 2} + \frac{1}{x - 1} \right] dx$$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + \ln|x - 1| + C$$

$$\int \frac{3x^2 + 1}{x^4 - 1} dx$$

$$\frac{3x^2 + 1}{(x^2 - 1)(x^2 + 1)}$$

$$\frac{3x^2 + 1}{(x-1)(x+1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

$$* (x-1)(x+1)(x^2+1)$$

$$3x^2 + 1 = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x-1)(x+1)$$

$$\underline{x=1}$$

$$3+1 = A(2)(2) + 0 + 0$$

$$4 = 4A \rightarrow A = 1$$

$$\underline{x=-1}$$

$$3+1 = 0 + B(-2)(2) + 0$$

$$4 = -4B \rightarrow B = -1$$

$$\underline{x=0}$$

$$0+1 = A(1)(1) + B(-1)(1) + D(-1)(1)$$

$$1 = A - B - D$$

$$1 = 1 - (-1) - D \rightarrow 1 = 2 - D$$

Equate the coeff. of x^2

$$D = 1$$

$$0 = A + B + C$$

$$0 = 1 - 1 + C \rightarrow C = 0$$

$$\int \left[\frac{1}{x-1} + \frac{-1}{x+1} + \frac{0x+1}{x^2+1} \right] dx$$

$$\ln|x-1| - \ln|x+1| + \tan^{-1}(x) + c$$

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$$

$$\frac{x^2 + 2x - 1}{x(2x^2 + 3x - 2)}$$

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

$$* \quad x(2x - 1)(x + 2)$$

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

$$\underline{x=0} \quad 0 + 0 - 1 = A(-1)(2) + 0 + 0$$

$$-1 = -2A \rightarrow A = \frac{1}{2}$$

$$\underline{x=\frac{1}{2}} \quad \frac{1}{4} + 1 - 1 = 0 + B\frac{1}{2}\left(\frac{1}{2} + 2\right) + 0$$

$$\frac{1}{4} = B\frac{1}{2}\left(\frac{5}{2}\right)$$

$$\frac{1}{4} = \frac{5}{4}B \rightarrow B = \frac{1}{5}$$

$$\underline{x=-2} \quad 4 - 4 - 1 = 0 + 0 + C(-2)(-4 - 1)$$

$$-1 = 10C \rightarrow C = -\frac{1}{10}$$

$$\int \left[\frac{\frac{1}{2}}{x} + \frac{\frac{1}{5}}{2x - 1} + \frac{-\frac{1}{10}}{x + 2} \right] dx$$

$$= \frac{1}{2} \ln|x| + \frac{1}{5} \ln|2x - 1| - \frac{1}{10} \ln|x + 2| + C$$

$$\int \frac{6x + 12}{x^4 + x^2 - 2} dx$$

$$\frac{6x + 12}{(x^2 - 1)(x^2 + 2)}$$

$$\frac{6x + 12}{(x-1)(x+1)(x^2+2)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+2}$$

$$* (x-1)(x+1)(x^2+2)$$

$$6x + 12 = A(x+1)(x^2+2) + B(x-1)(x^2+2) + (Cx+D)(x-1)(x+1)$$

$$\underline{x=1} \quad 6+12 = A(2)(3) + 0 + 0 \rightarrow A = \frac{18}{6}$$

$$\boxed{A = 3}$$

$$\underline{x=-1}$$

$$-6+12 = 0 + B(-2)(3) + 0 \rightarrow B = \frac{6}{-6}$$

$$\boxed{B = -1}$$

$$\underline{x=0}$$

$$12 = A(1)(2) + B(-1)(2) + D(-1)(1)$$

$$12 = 3(2) - 1(-2) - D$$

$$12 = 6 + 2$$

$$-D \rightarrow \boxed{D = -4}$$

Equate the coeff. of x^3

$$0 = A + B + C$$

$$0 = 3 - 1 + C \rightarrow \boxed{C = -2}$$

$$\int \left[\frac{3}{x-1} + \frac{-1}{x+1} + \frac{-2x-4}{x^2+2} \right] dx$$

$$3 \ln|x-1| - \ln|x+1| + \int \frac{-(2x)-4}{x^2+2} dx$$

$$\int \left(-\frac{(2x)}{x^2+2} + \frac{-4}{x^2+2} \right) dx$$

$$3 \ln|x-1| - \ln|x+1| - \ln|x^2+2| - 4 \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$$



Text Book Question



$$\int \frac{1}{x^3 - 1} dx$$

$$\frac{1}{(x-1)(x^2+x+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$$

$$* (x-1)(x^2+x+1)$$

$$1 = A(x^2+x+1) + (Bx+C)(x-1)$$

$$\underline{x=1} \quad 1 = A(3) + 0 \longrightarrow \boxed{A = \frac{1}{3}}$$

$$\underline{x=0} \quad 1 = A(1) + C(-1)$$

$$1 = \frac{1}{3} - C \longrightarrow C = \frac{1}{3} - 1 \longrightarrow \boxed{C = -\frac{2}{3}}$$

Equate the coeff of x^2

$$0 = A + B$$

$$0 = \frac{1}{3} + B$$

$$\longrightarrow \boxed{B = -\frac{1}{3}}$$

$$\int \left[\frac{\frac{1}{3}}{x-1} + \frac{-\frac{1}{3}x + \frac{-2}{3}}{x^2+x+1} \right] dx$$

$$\frac{1}{3} \ln|x-1| + \left(-\frac{1}{3}\right) \int \frac{x+1}{x^2+x+1} dx$$

$$\int \frac{\frac{1}{2}(2x+1) + \frac{1}{2}}{x^2+x+1} dx$$

$$\int \left[\frac{\frac{1}{2}(2x+1)}{x^2+x+1} + \frac{\frac{1}{2}}{x^2+x+1} \right] dx$$

$$\downarrow$$

$$(x + \frac{1}{2})^2 - \frac{1}{4} + 1$$

$$(x + \frac{1}{2})^2 + \frac{3}{4}$$

$$\frac{1}{2} \ln|x^2+x+1| + \frac{1}{2} \int \frac{1}{(x+\frac{1}{2})^2 + \frac{3}{4}} dx$$

$$\frac{1}{2} \ln|x^2+x+1| + \frac{1}{2} \cdot \frac{1}{\sqrt{3/4}} \tan^{-1}\left(\frac{x+\frac{1}{2}}{\sqrt{3/4}}\right) + C$$

$$\frac{1}{3} \ln|x-1| - \frac{1}{3} \left[\frac{1}{2} \ln|x^2+x+1| \right.$$

$$\left. + \frac{1}{2} \cdot \frac{1}{\sqrt{3/4}} \tan^{-1}\left(\frac{x+\frac{1}{2}}{\sqrt{3/4}}\right) + C \right]$$

$$\int \frac{u^2 + 3}{1 - u^4} du$$

$$- (u^4 - 1)$$

$$- \int \frac{u^2 + 3}{u^4 - 1} du$$

$$\frac{u^2 + 3}{(u^2 - 1)(u^2 + 1)}$$

$$\frac{u^2 + 3}{(u - 1)(u + 1)(u^2 + 1)} = \frac{A}{u - 1} + \frac{B}{u + 1} + \frac{Cu + D}{u^2 + 1}$$

$$* \underline{(u - 1)(u + 1)(u^2 + 1)}$$

$$u^2 + 3 = A(u + 1)(u^2 + 1) + B(u - 1)(u^2 + 1) + (Cu + D)(u - 1)(u + 1)$$

$$\underline{u = 1} \quad L = A(2)(2) + 0 + 0 \rightarrow \boxed{A = 1}$$

$$\underline{u = -1} \quad L = 0 + B(-2)(2) + 0 \rightarrow \boxed{B = -1}$$

$$\underline{u = 0} \quad 3 = A(1)(1) + B(-1)(1) + D(-1)(1)$$

$$3 = A + B - D$$

$$3 = 1 + 1 - D \rightarrow \boxed{D = -1}$$

Equate the coeff of u^3

$$0 = A + B + C$$

$$0 = 1 - 1 + C \rightarrow \boxed{C = 0}$$

$$= - \int \left[\frac{1}{u - 1} + \frac{-1}{u + 1} + \frac{0 - 1}{u^2 + 1} \right] du$$

$$= - \left[\ln|u - 1| - \ln|u + 1| - \tan^{-1} u \right] + c$$

$$\int \frac{1}{x^3 + 4x} dx$$

$$\frac{1}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

* $x(x^2+4)$

$$1 = A(x^2+4) + (Bx+C)x$$

$x=0$ $1 = A(4) + 0 \rightarrow A = \frac{1}{4}$

Equate the coeff of x^2

$$0 = A + B$$

$$0 = \frac{1}{4} + B \rightarrow B = -\frac{1}{4}$$

6.5.7

Equate the coeff of x

$$0 = C$$

$$\int \left[\frac{\frac{1}{4}}{x} + \frac{-\frac{1}{4}x + 0}{x^2+4} \right] dx$$

$$\frac{1}{4} \ln|x| - \frac{1}{4} \int \frac{x}{x^2+4} dx$$

$$\int \frac{\frac{1}{2}(2x)}{x^2+4} dx$$

$$\Rightarrow \frac{1}{4} \ln|x| - \frac{1}{4} \cdot \frac{1}{2} \ln|x^2+4| + C$$

$$\int \frac{2x^2 - 10x + 15}{x^3 - 4x^2 + 5x} dx$$

$$\frac{2x^2 - 10x + 15}{x(x^2 - 4x + 5)} = \frac{A}{x} + \frac{Bx + C}{x^2 - 4x + 5}$$

$$* x(x^2 - 4x + 5)$$

$$2x^2 - 10x + 15 = A(x^2 - 4x + 5) + (Bx + C)x$$

$$\underline{x=0} \quad 0 - 0 + 15 = A(5) + 0 \rightarrow \boxed{A=3}$$

Equate the coeff. of x^2

$$2 = A + B$$

$$2 = 3 + B \rightarrow \boxed{B=-1}$$

~~خطا~~

Equate the coeff. of x

$$-10 = -4A + C$$

$$-10 = -4(3) + C$$

$$-10 = -12 + C \rightarrow \boxed{C=2}$$

$$\int \left[\frac{3}{x} + \frac{-x + 2}{x^2 - 4x + 5} \right] dx$$

$$3 \ln|x| + \int \frac{-\frac{1}{2}(2x-4)}{x^2 - 4x + 5} dx$$

$$\Rightarrow 3 \ln|x| - \frac{1}{2} \ln|x^2 - 4x + 5| + C$$

$$\int \frac{3x^2 + x + 4}{x^4 + 3x^2 + 2} dx$$



$$\frac{3x^2 + x + 4}{(x^2 + 1)(x^2 + 2)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 2}$$

* $(x^2 + 1)(x^2 + 2)$

$$3x^2 + x + 4 = (Ax + B)(x^2 + 2) + (Cx + D)(x^2 + 1)$$

$x=0$ $0 + 0 + 4 = (0 + B)(0 + 2) + (0 + D)(0 + 1)$

$$4 = 2B + D \rightarrow D = 4 - 2B \quad (1)$$

Equate the coeff. of x^3 $0 = A + C \rightarrow C = -A \quad (2)$

Equate the coeff. of x^2 $3 = B + D \quad (3)$

Equate the coeff. of x $1 = 2A + C \quad (4)$

(1) in (3)

$$3 = B + 4 - 2B$$

$$3 = 4 - B \rightarrow B = 1$$

in (1) $D = 4 - 2 = 2$

(2) in (4)

$$1 = 2A - A \rightarrow A = 1$$

in (2) $C = -1$

$$\int \left[\frac{x+1}{x^2+1} + \frac{-x+2}{x^2+2} \right] dx$$

$$\int \left[\frac{x}{x^2+1} + \frac{1}{x^2+1} - \frac{x}{x^2+2} + \frac{2}{x^2+2} \right] dx$$

$$\frac{1}{2} \ln|x^2+1| + \tan^{-1} x - \frac{1}{2} \ln|x^2+2| + 2 \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x}{\sqrt{2}} \right) + c$$

يُعتبر البني الحل
على In بدل
توزيع المقام



Text Book Question



$$\int \frac{x^3 - 2x^2 + 2x - 5}{x^4 + 4x^2 + 3} dx$$

$$\frac{x^3 - 2x^2 + 2x - 5}{(x^2 + 1)(x^2 + 3)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 3}$$

$$* (x^2 + 1)(x^2 + 3)$$

$$x^3 - 2x^2 + 2x - 5 = (Ax + B)(x^2 + 3) + (Cx + D)(x^2 + 1)$$

$$\underline{x=0} \quad -5 = B(3) + D(1) \longrightarrow D = -5 - 3B \quad (1)$$

$$\underline{\text{Equate the coeff. of } x^3} \quad 1 = A + C \longrightarrow C = 1 - A \quad (2)$$

$$\underline{\text{Equate the coeff. of } x^2} \quad -2 = B + D \quad (3)$$

$$\underline{\text{Equate the coeff. of } x} \quad 2 = 3A + C \quad (4)$$

$$(1) \text{ in } (3) \quad -2 = B - 5 - 3B$$
$$-2 + 5 = -2B \longrightarrow 3 = -2B \longrightarrow \boxed{B = -\frac{3}{2}}$$

$$B = -\frac{3}{2} \xrightarrow{\text{in } (3)} -2 = \frac{-3}{2} + D \longrightarrow D = -2 + \frac{3}{2}$$

$$\boxed{D = -\frac{1}{2}}$$

$$(2) \text{ in } (4) \quad 2 = 3A + 1 - A \longrightarrow 1 = 2A \longrightarrow \boxed{A = \frac{1}{2}}$$

$$A = \frac{1}{2} \xrightarrow{\text{in } (2)} C = 1 - \frac{1}{2} \longrightarrow \boxed{C = \frac{1}{2}}$$

$$\int \frac{2}{2x^2 + 3x + 1} dx$$

$$2 \left[x^2 + \frac{3}{2}x + \frac{1}{2} \right]$$

$$2 \left[\left(x + \frac{3}{4} \right)^2 - \frac{9}{16} + \frac{1}{2} \right]$$

$$2 \left[\left(x + \frac{3}{4} \right)^2 - \frac{9}{16} + \frac{8}{16} \right]$$

$$2 \left[\left(x + \frac{3}{4} \right)^2 - \frac{1}{16} \right]$$



الأحلى



$$\int \frac{1}{2 \left[\left(x + \frac{3}{4} \right)^2 - \frac{1}{16} \right]} dx$$

$$\frac{1}{2} \frac{1}{2\sqrt{\frac{1}{16}}} \ln \left| \frac{x + \frac{3}{4} - \sqrt{\frac{1}{16}}}{x + \frac{3}{4} + \sqrt{\frac{1}{16}}} \right| + C$$

$$\int \frac{2}{2x^2 + 3x + 1} dx$$

Solution#2

المقام يتحلل



Partial fraction

$$\frac{2}{(2x+1)(x+1)} = \frac{A}{2x+1} + \frac{B}{x+1}$$

$$* (2x+1)(x+1)$$

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$$2 = A(x+1) + B(2x+1)$$

$$\underline{x = -1}$$

$$2 = 0 + B(-2+1) \longrightarrow 2 = -B \longrightarrow B = -2$$

$$\underline{x = -\frac{1}{2}}$$

$$2 = A\left(-\frac{1}{2}+1\right) \longrightarrow 2 = \frac{1}{2}A \longrightarrow A = 4$$

$$\int \left[\frac{4}{2x+1} + \frac{-2}{x+1} \right] dx$$

$$4 \int \frac{1}{2x+1} dx + \int \frac{-2}{x+1} dx$$

$$= 4 \cdot \frac{1}{2} \ln|2x+1| - 2 \ln|x+1| + C$$



Text Book Question



$$\int_0^1 \frac{x-4}{x^2-5x+6} dx$$

$$\frac{x-4}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

$$* (x-2)(x-3)$$

$$x-4 = A(x-3) + B(x-2)$$

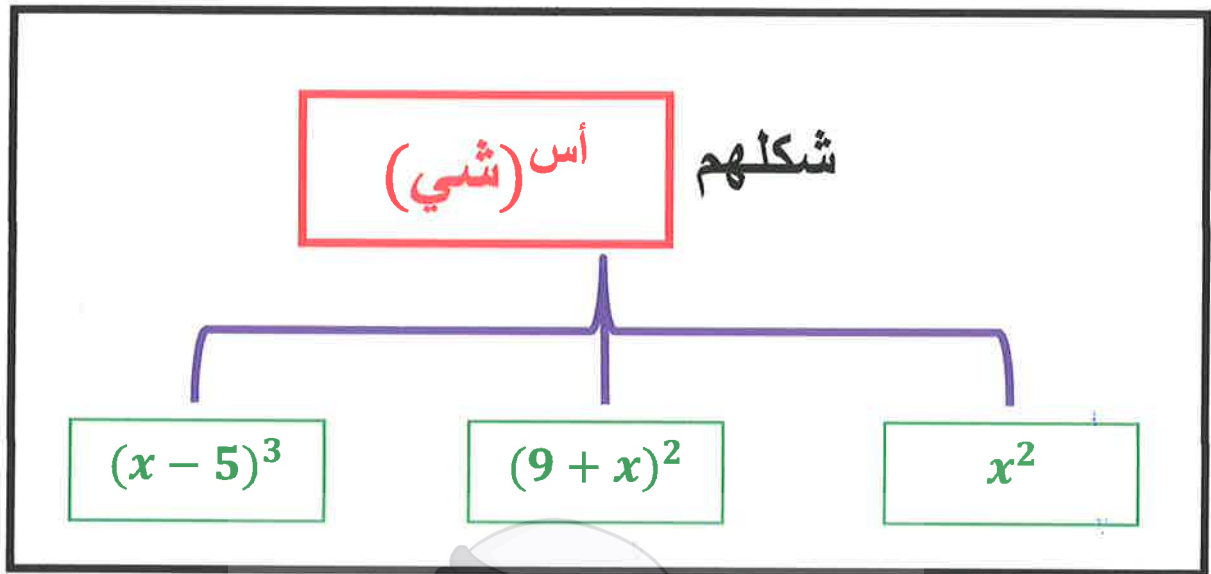
$$\underline{x=2} \quad -2 = A(-1) + 0 \rightarrow A = 2$$

$$\underline{x=3} \quad -1 = 0 + B(1) \rightarrow B = -1$$

$$\begin{aligned} & \int_0^1 \left[\frac{2}{x-2} + \frac{-1}{x-3} \right] dx \\ &= \left[2 \ln|x-2| - \ln|x-3| \right]_0^1 \\ &= (2 \ln|1-2| - \ln|1-3|) - (2 \ln|0-2| - \ln|0-3|) \\ &= 2 \ln 1 - \ln 2 - 2 \ln 2 + \ln 3 \\ &= -3 \ln 2 + \ln 3 \end{aligned}$$



الحدود المكررة



Examples

$$\int \frac{3x - 1}{(x + 3)^2(x - 2)} dx$$

$$\frac{3x - 1}{(x + 3)^2(x - 2)} = \frac{A}{x + 3} + \frac{B}{(x + 3)^2} + \frac{C}{x - 2}$$

$$\int \frac{3x - 1}{(x + 3)^3(x - 2)} dx$$

$$\frac{3x - 1}{(x + 3)^3(x - 2)} = \frac{A}{x + 3} + \frac{B}{(x + 3)^2} + \frac{C}{(x + 3)^3} + \frac{D}{x - 2}$$

$$\int \frac{x+1}{x^3-x^2} dx$$

$$\frac{x+1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$$

* $x^2(x-1)$

$$x+1 = A x(x-1) + B(x-1) + C x^2$$

$x=0$ $1 = 0 + B(-1) + 0 \rightarrow B = -1$

$x=1$ $2 = 0 + 0 + C \rightarrow C = 2$

Equate the coeff. of x^2

$$0 = A + C$$

$$0 = A + 2 \rightarrow A = -2$$

$$\int \left[\frac{-2}{x} + \frac{-1}{x^2} + \frac{2}{x-1} \right] dx$$

$-x^{-2}$

$$-2 \ln|x| - \frac{x^{-1}}{-1} + 2 \ln|x-1| + C$$

$$\int \frac{1}{x^4 - x^2} dx$$

$$\frac{1}{x^2(x^2-1)}$$

$$\frac{1}{x^2(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{x+1}$$

$$* \underline{x^2(x-1)(x+1)}$$

$$1 = A x(x-1)(x+1) + B(x-1)(x+1) + C x^2(x+1) + D x^2(x-1)$$

$$\underline{x=0}$$

$$1 = 0 + B(-1)(1) + 0 + 0 \rightarrow B = -1$$

$$\underline{x=1}$$

$$1 = 0 + 0 + C(2) + 0 \rightarrow C = \frac{1}{2}$$

$$\underline{x=-1}$$

$$1 = 0 + 0 + 0 + D(-2) \rightarrow D = -\frac{1}{2}$$

Equate the coeff. of x^3

$$0 = A + C + D$$

$$0 = A + \frac{1}{2} - \frac{1}{2} \rightarrow A = 0$$

$$\int \left[\frac{-1}{x^2} + \frac{\frac{1}{2}}{x-1} + \frac{-\frac{1}{2}}{x+1} \right] dx$$

$$-x^{-2}$$

$$= -\frac{x^{-1}}{-1} + \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$$

$$\int \frac{4x}{x^3 - x^2 - x + 1} dx$$

$$\frac{4x}{x^2(x-1) - (x-1)} = \frac{4x}{(x-1)[x^2-1]}$$

$$\frac{4x}{(x-1)(x-1)(x+1)}$$

$$\frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

$$* (x-1)^2(x+1)$$

$$4x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

$$\underline{x=1} \quad 4 = 0 + B(2) + 0 \rightarrow \boxed{B=2}$$

$$\underline{x=-1} \quad -4 = 0 + 0 + C(-2)^2 \rightarrow \boxed{C=-1}$$

$$\underline{x=0} \quad 0 = A(-1)(1) + B(1) + C(-1)^2$$

$$0 = -A + 2 - 1 \rightarrow \boxed{A=1}$$

$$\int \left[\frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{-1}{x+1} \right] dx$$

$$2(x-1)^{-2}$$

$$= \ln|x-1| + 2 \frac{(x-1)^{-1}}{-1} - \ln|x+1| + C$$

$$\int \frac{x^5}{(x^2 - 4)(x^2 + 4)} dx$$

درجة البسط
ك
درجة المقام
↓
قسمة طويلة

$$\int \frac{x^5}{x^4 - 16} dx$$

$$\begin{array}{r} x \\ -16 + x^4 \overline{) x^5} \\ \underline{-(x^4 - 16x)} \\ 16x \end{array}$$

$$\int \left[x + \frac{16x}{x^4 - 16} \right] dx$$

$$\left(\frac{x^2}{2} \right)$$

لا تنسوا
أخرى

$$+ \int \frac{16x}{x^4 - 16} dx$$

حل عجيب

$$\int \frac{16x}{(x^2)^2 - 16} dx$$

$$u = x^2$$

$$du = 2x dx \rightarrow \frac{1}{2} du = x dx$$

$$\frac{16}{2} \int \frac{du}{u^2 - 16}$$

$$8 \frac{1}{204} \ln \left| \frac{u-4}{u+4} \right| + C$$

$x^2 \leftarrow \frac{u-4}{u+4}$

$$\frac{16x}{(x^2 - 4)(x^2 + 4)}$$

$$\frac{16x}{(x-2)(x+2)(x^2+4)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{Cx+D}{x^2+4}$$

$$* (x-2)(x+2)(x^2+4)$$



See Next Page Plz!

$$16x = A(x+2)(x^2+4) + B(x-2)(x^2+4)$$

$$(Cx + D)(x-2)(x+2)$$

$$\underline{x=2}$$

$$32 = A(4)(8) + 0 + 0 \rightarrow \boxed{A=1}$$

$$\underline{x=-2}$$

$$-32 = 0 + B(-4)(8) + 0 \rightarrow \boxed{B=1}$$

$$\underline{x=0}$$

$$0 = A(2)(4) + B(-2)(4) + D(-2)(2)$$

$$0 = 8A - 8B - 4D$$

$$0 = \cancel{8} - \cancel{8} - 4D \rightarrow \boxed{D=0}$$

Equate the coeff. of x^3

$$0 = A + B + C$$

$$0 = 1 + 1 + C \rightarrow \boxed{C=-2}$$

$$\int \left[\frac{1}{x-2} + \frac{1}{x+2} + \frac{-2x+0}{x^2+4} \right] dx$$

$$= \ln|x-2| + \ln|x+2| - \ln|x^2+4| + C$$

$$\Rightarrow \left(\frac{x^2}{2} \right) + \ln|x-2| + \ln|x+2| - \ln|x^2+4| + C$$



Text Book Question



$$\int x \tan^{-1} x dx$$

$$u = \tan^{-1} x$$

$$dv = x dx$$

$$du = \frac{1}{x^2+1} dx$$

$$v = \frac{x^2}{2}$$

$$\frac{1}{2} \tan^{-1} x \cdot x^2 - \frac{1}{2} \int \frac{x^2}{x^2+1} dx$$

$$\begin{array}{r} 1 \\ 1+x^2 \overline{) x^2} \\ \underline{-x^2} \\ 1 \end{array}$$

$$\int \left[1 + \frac{-1}{x^2+1} \right] dx$$

$$x - \tan^{-1} x + c$$

$$\frac{1}{2} \tan^{-1} x \cdot x^2 - \frac{1}{2} [x - \tan^{-1} x] + c$$

$$\int \ln(x^2 + 2x + 2) dx$$

$$u = \ln(x^2 + 2x + 2)$$

$$dv = dx$$

$$du = \frac{2x + 2}{x^2 + 2x + 2} dx$$

$$v = x$$

$$\ln(x^2 + 2x + 2) \cdot x - \int x \frac{2x + 2}{x^2 + 2x + 2} dx$$

$$\int \frac{2x^2 + 2x}{x^2 + 2x + 2} dx$$

$$\begin{array}{r} 2 + 2x + x^2 \overline{) 2x^2 + 2x} \\ \underline{- 2x^2 + 4x + 4} \end{array}$$

$$\int \left[2 + \frac{-2x - 4}{x^2 + 2x + 2} \right] dx$$

$$2x + \int \frac{-(2x + 2) - 2}{x^2 + 2x + 2} dx$$

$$\int \left[\frac{-(2x + 2)}{x^2 + 2x + 2} + \frac{-2}{x^2 + 2x + 2} \right] dx$$

$$- \ln|x^2 + 2x + 2| + \int \frac{-2}{x^2 + 2x + 2} dx$$

$$(x + 1)^2 - 1 + 2$$

$$\int \frac{-2}{(x + 1)^2 + 1} dx$$

$$- 2 \tan^{-1}(x + 1) + C$$

$$\ln(x^2 + 2x + 2) x - \left[2x - \ln|x^2 + 2x + 2| - 2 \tan^{-1}(x + 1) \right] + C$$

$$\int \ln(x^2 + 2x + 10) dx$$

Oct 2017

$$u = \ln(x^2 + 2x + 10) \quad dv = dx$$

$$du = \frac{2x+2}{x^2+2x+10} dx \quad v = x$$

$$\ln(x^2+2x+10) \cdot x - \int x \frac{2x+2}{x^2+2x+10} dx$$

$$\int \frac{2x^2+2x}{x^2+2x+10} dx$$

$$\begin{array}{r} 2 \\ 10+2x+x^2 \overline{) 2x^2+2x} \\ \underline{2x^2+4x+20} \\ -2x-20 \end{array}$$

$$\int \left[2 + \frac{-2x-20}{x^2+2x+10} \right] dx$$

$$2x + \int \frac{-(2x+2)-18}{x^2+2x+10} dx$$

$$\int \left[\frac{-(2x+2)}{x^2+2x+10} + \frac{-18}{x^2+2x+10} \right] dx$$

$$- \ln|x^2+2x+10| + \int \frac{-18}{x^2+2x+10} dx$$

$$\ln(x^2+2x+10)x - \left[2x - \ln|x^2+2x+10| - 6 \tan^{-1}\left(\frac{x+1}{3}\right) \right] + C$$

$$-18 \cdot \frac{1}{3} \tan^{-1}\left(\frac{x+1}{3}\right) + C$$

What is the form of the partial fraction decomposition of

$$f(x) = \frac{2x}{x^2 - 1}$$

$$\frac{2x}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$$



Write out the form of the partial decomposition of the function

$$f(x) = \frac{2x^2 + 3x + 2}{x^3 + x^2 + 2x + 2}$$

$$\frac{2x^2 + 3x + 2}{x^2(x+1) + 2(x+1)}$$

$$\frac{2x^2 + 3x + 2}{(x+1)[x^2 + 2]} = \frac{A}{x+1} + \frac{Bx + C}{x^2 + 2}$$

$$f(x) = \frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3}$$

$$= \frac{A}{x} + \frac{B}{x-1} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{x^2 + 1} + \frac{Gx + H}{(x^2 + 1)^2} + \frac{Ix + J}{(x^2 + 1)^3}$$



Multiple choice



The partial fraction decomposition of $\frac{x^2 - 6}{x^3 + 8}$ is

(a) $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$

(b) $\frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

(c) $\frac{A}{x+2} + \frac{Bx+C}{x^2-2x+4}$

(d) $\frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$

(e) $\frac{A}{x+2} + \frac{Bx+C}{x^2+2x+4}$

Oct 2017

$$\frac{x^2 - 6}{x^3 + 8} = \frac{x^2 - 6}{(x+2)(x^2 - 2x + 4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2 - 2x + 4}$$

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طريقة التفكير بالتكامل في الامتحان



التبسيط قبل الحل

هناك دالة بالمقلم مشتقتها
موجود بالبسط

زاوية أو أس ليس حدودية
من الدرجة الأولى

ثم

By parts

عند وجود أولوية
من الأولويات

بس دوال مثلثية
أو (h)

حدودية من
الدرجة الثانية

دوال مثلثية بالمقام
وما تصعد بالبسط

حدودية من
الدرجة الثانية $\frac{1}{2} \frac{3}{2} \frac{5}{2} \dots$

حدودية
حدودية

المقام < البسط

Partial fraction

البسط $\leq$ المقام

قسمة مطولة



Text Book Question



$$\int \frac{e^x}{(e^x - 2)(e^{2x} + 1)} dx$$

Let $u = e^x$
 $du = e^x dx$

$$\int \frac{1}{(u-2)(u^2+1)} du$$

$$\frac{1}{(u-2)(u^2+1)} = \frac{A}{u-2} + \frac{Bu+C}{u^2+1}$$

* $(u-2)(u^2+1)$

$$1 = A(u^2+1) + (Bu+C)(u-2)$$

$u=2$

$$1 = A(4+1) + 0 \rightarrow A = \frac{1}{5}$$

$u=0$

$$1 = A(1) + (C)(-2)$$

$$1 = \frac{1}{5} - 2C$$

$$2C = \frac{1}{5} - 1 \rightarrow 2C = -\frac{4}{5}$$

$$C = \frac{1}{2} \left(-\frac{4}{5}\right)$$

$$C = -\frac{2}{5}$$

Equate the coeff. of u^2

$$0 = A + B$$

$$0 = \frac{1}{5} + B \rightarrow B = -\frac{1}{5}$$

تم اكمال الحل



Text Book Question



$$\int \frac{e^{2x}}{e^{2x} + 3e^x + 2} dx$$

Let $u = e^x$
 $du = e^x dx$

$$\int \frac{e^x \cdot e^x}{(e^x)^2 + 3e^x + 2} dx$$

$$\int \frac{u}{u^2 + 3u + 2} du \Rightarrow$$

SOL#1

SOL#2
 يمكن بناء الحل
 على
 ln

$$\frac{u}{(u+1)(u+2)} = \frac{A}{u+1} + \frac{B}{u+2}$$

* $(u+1)(u+2)$

$$u = A(u+2) + B(u+1)$$

$u = -2$ $-2 = 0 + B(-1) \rightarrow \boxed{B = 2}$

$u = -1$ $-1 = A(1) + 0 \rightarrow \boxed{A = -1}$

$$\int \left[\frac{-1}{u+1} + \frac{2}{u+2} \right] du$$

$$= -\ln|u+1| + 2\ln|u+2| + C$$

$$= -\ln|e^x+1| + 2\ln|e^x+2| + C$$



Text Book Question



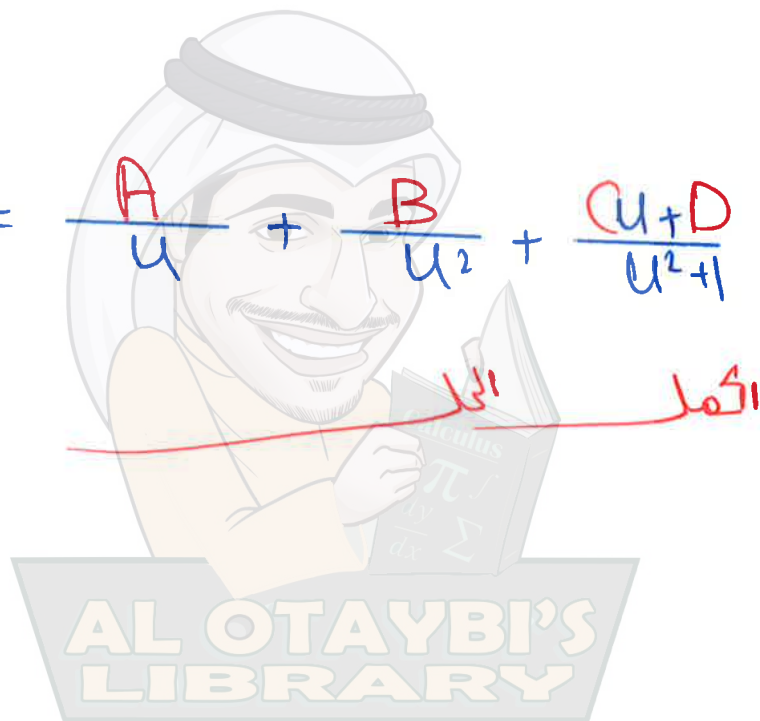
$$\int \frac{\cosh t}{\sinh^2 t + \sinh^4 t} dt$$

Let $u = \sinh t$
 $du = \cosh t \, dt$

$$\int \frac{1}{u^2 + u^4} du$$

$$\frac{1}{u^2(1+u^2)}$$

$$\frac{1}{u^2(u^2+1)} = \frac{A}{u} + \frac{B}{u^2} + \frac{Cu+D}{u^2+1}$$



$$\int \frac{\sec^2 x}{\tan^2 x + 3 \tan x + 2} dx$$

Let $u = \tan x$
 $du = \sec^2 x dx$

$$\int \frac{1}{u^2 + 3u + 2} du$$

SOL #1

$$u^2 + 3u + 2$$

$$(u + \frac{3}{2})^2 - \frac{9}{4} + 2$$

$$(u + \frac{3}{2})^2 - \frac{9}{4} + \frac{8}{4}$$

$$(u + \frac{3}{2})^2 - \frac{1}{4}$$

$$\int \frac{1}{(u + \frac{3}{2})^2 - \frac{1}{4}} du$$

$$\ln \left| \frac{u + \frac{3}{2} - \frac{1}{2}}{u + \frac{3}{2} + \frac{1}{2}} \right| + C$$

$$\ln \left| \frac{\tan x - \frac{2}{2}}{\tan x + \frac{4}{2}} \right| + C$$

SOL #2 لأن المقام يتحلل

$$\frac{1}{(u+1)(u+2)} = \frac{A}{u+1} + \frac{B}{u+2}$$

$$* (u+1)(u+2)$$

$$1 = A(u+2) + B(u+1)$$

$$\underline{u = -1} \quad 1 = A(1) + 0 \rightarrow \boxed{A = 1}$$

$$\underline{u = -2} \quad 1 = 0 + B(-1) \rightarrow \boxed{B = -1}$$

$$\int \left[\frac{1}{u+1} + \frac{-1}{u+2} \right] du$$

$$\ln|u+1| - \ln|u+2| + C$$

$$\ln|\tan x + 1| - \ln|\tan x + 2| + C$$

$$\int \frac{\sin x}{\cos^2 x - 3\cos x} dx$$

Let $u = \cos x$

$du = -\sin x dx \longrightarrow -du = \sin x dx$

$$- \int \frac{1}{u^2 - 3u} du$$

→ Sol #1

Sol #2 لأن المقام يتكامل

$$\frac{1}{u(u-3)} = \frac{A}{u} + \frac{B}{u-3}$$

* $u(u-3)$

$$1 = A(u-3) + Bu$$

$u=3$

$$1 = 0 + 3B \rightarrow B = \frac{1}{3}$$

$u=0$

$$1 = A(-3) + 0 \rightarrow A = -\frac{1}{3}$$

$$- \int \left[\frac{-\frac{1}{3}}{u} + \frac{\frac{1}{3}}{u-3} \right] du$$

$$- \left[\frac{-\frac{1}{3}}{3} \ln|u| + \frac{\frac{1}{3}}{3} \ln|u-3| \right] + C$$

$$- \left[-\frac{1}{3} \ln|\cos x| + \frac{1}{3} \ln|\cos x - 3| \right] + C$$

$$u^2 - 3u$$

$$\left(u - \frac{3}{2}\right)^2 - \frac{9}{4}$$

$$- \int \frac{1}{\left(u - \frac{3}{2}\right)^2 - \frac{9}{4}} du$$

$$- \ln \left| \frac{u - \frac{3}{2} - \frac{3}{2}}{u - \frac{3}{2} + \frac{3}{2}} \right| + C$$

$$- \ln \left| \frac{\cos x - \frac{6}{4}}{\cos x} \right| + C$$



Text Book Question



$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{2 + \cos x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{2 \sin x \cos x}{2 + \cos x} dx$$

Let $u = \cos x$

$$du = -\sin x dx \rightarrow -du = \sin x dx$$

$$- \int \frac{2u}{2+u} du$$

$$- \int \left[2 + \frac{-4}{2+u} \right] du$$

$$\begin{array}{r} 2 \\ \hline 2u \\ -2u + 4 \\ \hline -4 \end{array}$$

$$- [2u - 4 \ln |2+u|]$$

$$- [2 \cos x - 4 \ln |2 + \cos x|]_0^{\frac{\pi}{2}}$$

$$- [0 - 4 \ln |2+0| - (2 - 4 \ln |2+1|)]$$

$$- [-4 \ln 2 - 2 + 4 \ln 3]$$

$$\int \frac{1}{(1+e^x)^2} dx$$

Let $u = e^x$
 $du = e^x dx$

$$\int \frac{1}{(1+e^x)^2} dx$$

$\frac{e^x}{e^x}$

$$\int \frac{e^x}{e^x (1+e^x)^2} dx$$

$$\int \frac{1}{u(1+u)^2} du$$

$$\frac{1}{u(u+1)^2} = \frac{A}{u} + \frac{B}{u+1} + \frac{C}{(u+1)^2}$$

* $u(u+1)^2$

$$1 = A(u+1)^2 + B u(u+1) + C u$$

$\downarrow (u^2 + 2u + 1)$

$u=0$ $1 = A(1) + 0 + 0 \rightarrow A = 1$

$u=-1$ $1 = 0 + 0 + C(-1) \rightarrow C = -1$

Equate the coeff. of u^2

$$0 = A + B$$

$$0 = 1 + B \rightarrow B = -1$$

$$\int \left[\frac{1}{u} + \frac{-1}{u+1} + \frac{-1}{(u+1)^2} \right] du$$

$\downarrow (u+1)^{-2}$

$$= \ln|u| - \ln|u+1| + \frac{(u+1)^{-1}}{-1} + C$$

$\downarrow e^x$

$$\int \frac{\ln 2}{2^x - 3(2^{-x}) + 2} dx$$

Let $u = 2^x$

$du = 2^x \ln 2 dx$

$$\int \frac{\ln 2}{2^x - 3(2^{-x}) + 2} dx$$

$\frac{2^x}{2^x}$

$$\int \frac{2^x \ln 2}{(2^x)^2 - 3(1) + 2(2^x)} dx$$

$$\int \frac{1}{u^2 + 2u - 3} du$$

Sol #1

$$u^2 + 2u - 3$$

$$(u+1)^2 - 1 - 3$$

$$(u+1)^2 - 4$$

$$\int \frac{1}{(u+1)^2 - 4} du$$

$$= \frac{1}{2(2)} \ln \left| \frac{u+1-2}{u+1+2} \right| + C$$

$$= \frac{1}{4} \ln \left| \frac{2^x - 1}{2^x + 3} \right| + C$$

Sol #2

$$\frac{1}{(u-1)(u+3)} = \frac{A}{u-1} + \frac{B}{u+3}$$

$$* (u-1)(u+3)$$

$$1 = A(u+3) + B(u-1)$$

$$u=1 \quad 1 = A(4) + 0 \rightarrow A = \frac{1}{4}$$

$$u=-3 \quad 1 = 0 + B(-4) \rightarrow B = -\frac{1}{4}$$

$$\int \left[\frac{\frac{1}{4}}{u-1} + \frac{-\frac{1}{4}}{u+3} \right] du$$

$$= \frac{1}{4} \ln |u-1| - \frac{1}{4} \ln |u+3| + C$$

$\xrightarrow{2^x} \quad \xrightarrow{2^x}$

$$\int \frac{\sinh x}{\sinh^2 x + 3 \cosh x + 3} dx$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\cosh^2 x - 1 = \sinh^2 x$$

$$\int \frac{\sinh x}{\cosh^2 x - 1 + 3 \cosh x + 3} dx$$

$$\int \frac{\sinh x}{\cosh^2 x + 3 \cosh x + 2} dx$$

Let $u = \cosh x$
 $du = \sinh x dx$

$$\int \frac{1}{u^2 + 3u + 2} du$$

SOL #1

$$u^2 + 3u + 2 = \left(u + \frac{3}{2}\right)^2 - \frac{9}{4} + 2 = \left(u + \frac{3}{2}\right)^2 - \frac{1}{4}$$

$$\int \frac{1}{\left(u + \frac{3}{2}\right)^2 - \frac{1}{4}} du$$

$$= \frac{1}{2\sqrt{\frac{1}{4}}} \ln \left| \frac{u + \frac{3}{2} - \frac{1}{2}}{u + \frac{3}{2} + \frac{1}{2}} \right| + C$$

$\cosh x$

SOL #2

المقام يتكامل

$$\frac{1}{(u+1)(u+2)} = \frac{A}{u+1} + \frac{B}{u+2}$$

$$* (u+1)(u+2)$$

$$1 = A(u+2) + B(u+1)$$

$$u = -1 \quad 1 = A(1) + 0 \rightarrow A = 1$$

$$u = -2 \quad 1 = 0 + B(-1) \rightarrow B = -1$$

$$\int \left[\frac{1}{u+1} + \frac{-1}{u+2} \right] du = \ln|u+1| - \ln|u+2| + C$$

$\cosh x$



Calculus B

Quiz # 5



$$\int \left[\frac{1}{x^2 + 5} + \frac{1}{x^2 - 5} \right] dx = \frac{1}{\sqrt{5}} \tan^{-1} \left(\frac{x}{\sqrt{5}} \right) + \frac{1}{2\sqrt{5}} \ln \left| \frac{x-\sqrt{5}}{x+\sqrt{5}} \right| + c$$

$$\int \frac{x^3}{x^4 + 2} dx = \left(\frac{1}{4} \right) \int \frac{4x^3}{x^4 + 2} dx = \frac{1}{4} \ln |x^4 + 2| + c$$

$$\int \frac{\cos x}{\sin^2 x + 9} dx = \int \frac{\cos x}{(\sin x)^2 + 9} dx \quad \left| \quad \int \frac{1}{u^2 + 9} du \right.$$

Let $u = \sin x$
 $du = \cos x dx$

$$= \frac{1}{3} \tan^{-1} \left(\frac{u}{3} \right) + c$$

$$= \frac{1}{3} \tan^{-1} \left(\frac{\sin x}{3} \right) + c$$

$$\int \frac{e^x}{e^{2x} - 9} dx = \int \frac{e^x}{(e^x)^2 - 9} dx \quad \left| \quad \int \frac{1}{u^2 - 9} du \right.$$

Let $u = e^x$
 $du = e^x dx$

$$= \frac{1}{2(3)} \ln \left| \frac{u-3}{u+3} \right| + c$$

$$= \frac{1}{6} \ln \left| \frac{e^x - 3}{e^x + 3} \right| + c$$

$$\int \frac{x}{x^4 + 9} dx = \int \frac{x}{(x^2)^2 + 9} dx \quad \left| \quad \frac{1}{2} \int \frac{1}{u^2 + 9} du \right.$$

Let $u = x^2$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$

$$\frac{1}{2} \cdot \frac{1}{3} \tan^{-1} \left(\frac{u}{3} \right) + c$$

$$\frac{1}{6} \tan^{-1} \left(\frac{x^2}{3} \right) + c$$

$$\int \frac{x^4}{x^2 - 9} dx$$

$$\begin{array}{r} x^2 + 9 \\ -9 + x^2 \overline{) x^4} \\ \underline{-x^4 + 9x^2} \\ 9x^2 \\ \underline{-9x^2 + 81} \\ 81 \end{array}$$

÷ ①
× ②
غير ③
+ ④

$$\int \left(x^2 + 9 + \frac{81}{x^2 - 9} \right) dx$$

$$= \frac{x^3}{3} + 9x + 81 \cdot \frac{1}{2(3)} \ln \left| \frac{x-3}{x+3} \right| + c$$